# Alexander (Sasha) Apartsin, Ph.D.


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## SELECTED RECOGNITION & ACHIEVEMENTS

**AI Product of the Year** recognition for SafeRide automotive AI product work.

**IIA innovation grants** for Motorola Solutions industry-academia AI projects.

**$250K Citi LabLink grant** for a financial data science collaboration platform.

**$10M hardware-cost savings contribution** through model-based capacity planning at Amdocs.

## INNOVATION & ENTREPRENEURSHIP

**Founder / CTO experience** across search, multimedia, cloud optimization, and applied AI ventures; corporate innovation through AI labs, grants, patents, and prototypes.

## RESEARCH & IMPACT

**Publications:** 40+ Scholar-indexed works and 470+ citations across ML, vision, NLP, deep learning, and signal processing.

**Patents:** 10 granted/filed across vehicle diagnostics, driver monitoring, model configuration, onboard inference, public-safety audio, and search/image matching.

## TEACHING & MENTORSHIP

**24+ supervised B.Sc. / M.Sc. final projects** since 2012, with teaching across HIT, Afeka, TAU, and BIU; multi-course, project-based series in Language, Vision, and Scalable AI, each with a companion online textbook, delivered over 16 past offerings with 112+ student course projects.

## AI/ML DOMAINS

Materials Discovery · Time Series · Recommender Systems · Anomaly Detection · Task Allocation · Text / NLP · Diffusion · Big Data · Speech · Vision · Audio · LLMs · Agents · Knowledge Graphs

## INDUSTRIES

Automotive · Predictive Maintenance · Finance · Public Safety · Healthcare · Telecom · Wireless · Media / Multimedia · Advertising · Document AI · EdTech · Search

## HANDS-ON

Python · PyTorch · Hugging Face · SparkNLP / spaCy · OpenCV · LangChain / LangGraph · OpenAI · C / C++ · SQL · Pandas / NumPy · Databricks · MLflow · Horovod · AWS / Azure

*AI researcher and technology leader with 30+ years across language, vision, signals, and generative AI.*

- **Researcher.** Work across computer vision, text and speech processing, signal processing, generative AI, recommender systems, and big data.
- **Innovator.** Design and define AI products with customers: shape problem framing and translate findings into intellectual property, systems, and product opportunities.
- **Builder.** Build systems and teams; manage cross-functional data-science and engineering groups while staying hands-on.
- **Educator.** Teach graduate LLM and GenAI courses, design curricula, and supervise B.Sc. / M.Sc. projects.

## LEADERSHIP & RESEARCH EXPERIENCE

### Faculty Member - HIT, School of Computer Science — 2024-Present

Conduct applied AI research in generative AI, large language models, vision, and signals; created hands-on, project-based course series on LLMs, GenAI, and diffusion models taught across HIT (School of Computer Science and Digital Healthcare Department) and Bar-Ilan University (undergraduate and graduate programs); supervise M.Sc. students in Afeka College of Engineering's Intelligent Systems program.

### Head of Data Science - Questar / SafeRide Technologies — 2020-2024

Built and hired the data science team; led big-data AI for vehicle telematics across signal, audio, graph, anomaly detection, and onboard-model work; led OEM-partner projects including engine-health monitoring (Ford), steering-by-wire monitoring (Hyundai Mobis), in-cabin acoustic fault detection (Renault), fuel-consumption modeling (Aptiv), and fault-sequence analysis (M.A.N.); engaged directly with customers to shape requirements; led the definition of a no-code product for onboard monitoring models.

### VP of Artificial Intelligence - harmon.ie — 2019-2020

Led AI for an enterprise document-management product designed to help organizations handle the growing amount of information in Office 365; managed a cross-functional data-science and engineering team and shaped the research agenda across large-scale text understanding, user-behavior modeling from Office-graph signals, and recommendation under limited supervision.

### Head of AI Research - Motorola Solutions, Israel Innovation Center — 2015-2019

Built and led the research team at Motorola Solutions' first AI lab in the Innovation Center; designed and led execution of the roadmap across video, text, speech, and public-safety AI; established internal cross-functional partnerships and academia-industry collaborations funded by Israel Innovation Authority programs and binational grants.

### AVP Data Science - Citi Israel Innovation Lab — 2013-2015

Led financial data-science projects: bond-price prediction for auctions, FX causality analysis and trader-informativeness classification, anomaly detection for trader compliance auditing, and trade-to-wire matching via correlation and constraint optimization.

### Founder / CTO / R&D and algorithm leadership - SearchNG, CloudTuner, MPCode, Tegrity, AppiVision, Amdocs, Comverse, earlier roles — 1991-2012

Founded ventures and led hands-on intelligent-system work in search, multimedia, video streaming, wireless communications, telecom, predictive modeling, handwriting recognition, imaging, inspection, and signal acquisition.

## EDUCATION

**Postdoctoral Fellowship** - Weizmann Institute of Science · computer vision and machine learning — 2013

**Ph.D., Computer Science** - Tel Aviv University · machine learning and signal processing — 2012

**M.Sc. in Management** - Polytechnic Institute of NYU · technology management — 2006

**M.Sc., Computer Science** - Weizmann Institute of Science · computer vision — 2004

**B.Sc., Computer Science** - Technion — 1995